Speech: slides 8–9

Part A, points (d)–(e)

About 2 minutes 15 seconds. *[red brackets]* = what to point at or do; // = a short pause. If Part A is drawn and you present it alone, open with slides 1–7 (separate PDF).

Slide 8 — A spillover of 1/(sigma minus 1) is Helpman in disguise

about 1 min 15 s

> *[title]* Point (d): Helpman 1998, in steps. In Helpman each location produces many varieties. Free entry makes their number proportional to population, n-i equals kappa L-i, and each variety sells at a constant markup over marginal cost.
>
> *[formula in the white card]* Step one: the contribution of location i to the price index of n is the number of varieties times their price to the one minus sigma. Since the number of varieties is proportional to L, that is **L-i to the power one**, times a term without L. Step two: in Armington with the spillover the price is w over A-bar L to the lambda, so the contribution is **L-i to the lambda times sigma minus one**, times a term without L.
>
> *[formula lambda = 1/(sigma-1)]* Step three: // the two coincide when lambda times sigma minus one equals one. So **lambda equals one over sigma minus one**. *[the two cards]* With sigma equal to 5, the value in ARM1, that is 0.25 — the value we used in points (b) and (c). With sigma equal to 4, the value in ARM2 and in Redding and Sturm, it is **one third**: keep this number in mind for Part B.
>
> *[bottom box, look at the Professor]* Step four: an estimated lambda need not be a technological externality, like knowledge spillovers. It can be the **variety gains** of monopolistic competition: more people, more firms, more varieties, a lower price index. Same equations, different stories — so the interpretation of lambda depends on the model behind it.

Slide 9 — What would identify lambda?

about 1 min

> *[title]* Point (e). If one cross-section cannot identify lambda, because the fundamentals adjust to rationalise any value, we need **variation over time**: a panel.
>
> *[red card, "What must change"]* What must change? **Population**, or the density of employment around a location, for reasons unrelated to productivity: a border that closes, a country divided, a new infrastructure.
>
> *[blue card, "What must stay fixed"]* What must stay fixed? The **fundamentals**: the shock must move people without changing what each place intrinsically is.

[formula] Then we invert the model at both dates. // With A-bar fixed, the change in log A is lambda times the change in log L, and that relation identifies lambda.

[bottom box] This is exactly what the next two parts do: the division of Germany shocks market access across cities, and the Berlin Wall takes away the employment around the blocks near the old centre without changing what those blocks are.

[hand over] My colleague will now present Part B.

Numbers to remember. Helpman: exponent 1; Armington: $\lambda(\sigma-1)$ · $\lambda = 1/(\sigma-1)$: 0.25 at $\sigma = 5$ (ARM1), 1/3 at $\sigma = 4$ (ARM2, Redding–Sturm) · $\Delta \log A = \lambda \Delta \log L$ with $\bar{A}$ fixed.
Words to get right. *one over sigma minus one*, never "one over one minus sigma" · $\sigma = 5$ vs 4 is ARM1 vs ARM2, not Armington vs Helpman · the shock moves *population*; productivity moves as a consequence.